RESEARCH & PROTOTYPE REVIEW

Business & Revenue Model Report

User segments, value hypotheses, monetisation options, unit economics and validation roadmap.

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SOURCE-CODE REVIEW + PUBLIC RESEARCH | 20 OCTOBER 2026

ALCHEMY ARCHITECTS

Executive position

AAKAASH ADHINAYAK can be positioned as an early research prototype for transparent, human-reviewed scheduling and data-quality workflows. At this stage, customer demand, willingness to pay, procurement path, support cost, and recurring revenue are not evidenced. The viable route is to validate a low-risk educational or civilian resource-planning use case before making broader claims.

Commercialisation status: HYPOTHESES ONLY. No customer, paid pilot, procurement contract, verified savings result or revenue has been established by the supplied files.

1. User segments and value hypotheses

Segment	Potential problem to test	Possible value proposition	Validation evidence
Engineering / logistics education	Students need reproducible exercises in constraints, data quality and disruption handling.	Instructor-ready sandbox, synthetic datasets, visible validation and report export.	Pilot with class, completion time, instructor feedback, repeat use.
Civilian airport / facility operations research	Schedules depend on changing capacity, maintenance and weather information.	Generic planning assistant for approved civilian tasks, with provenance and audit.	Workflow mapping, permissioned data, baseline comparison and user acceptance.
Inventory / maintenance teams	Priorities, availability, maintenance windows and data conflicts must be reconciled.	Inventory/task scheduling with a transparent rule engine and error explanations.	Shadow-mode pilot, constraint violation rate, staff hours, incident/error counts.
Research / consultancy partners	Need a controlled way to benchmark planning methods.	Scenario generation, reproducible metrics, solver comparison and traceable exports.	Research agreement, replicable benchmark and published method.

2. Possible business models

Model	How it could work	What must be true first	Risk
Education licence	Annual per-institution access to course material, exercises and sandbox.	Repeat use, instructor adoption, data rights, accessible onboarding.	Low willingness to pay without curriculum/instructor value.
Hosted subscription	Cloud-hosted generic planning/workflow tools with account tiers.	Secure multi-user deployment, stable uptime, privacy and measured value.	Hosting/support/API costs may exceed subscription margin.
Private deployment / integration	Configure adapters and schemas for approved civilian workflows.	Clear scope, customer-owned data approvals, testing and support capability.	Long integration cycles and maintenance burden.
Training / workshops	Paid sessions on data provenance, scheduling constraints and audit evidence.	Credible instructors, repeatable materials and proof of outcomes.	Labour-intensive; limited scalability.
Research grants / partnerships	Fund dataset access, evaluation studies and prototypes.	Research question, legal/data agreements, reproducibility.	Grant cycles are irregular and not recurring product revenue.

3. Illustrative pricing experiments - not a forecast

Any amount below is only a hypothesis for customer discovery in India, not a current market benchmark or a price recommendation based on validated demand. Discuss willingness to pay before publishing prices.

Offer to test	Hypothesis range	What to learn
Department/classroom pilot (time-limited)	INR 0-25,000 one-time	Can a faculty member run a class/demo without extensive support?

Offer to test	Hypothesis range	What to learn
Annual education licence	INR 15,000-1,00,000 per institution/year	What value do instructors need: scenarios, analytics, assessments, hosting or support?
Civilian workflow discovery workshop	INR 10,000-50,000 per engagement	Does the client pay for workflow discovery, benchmark design and requirements mapping?
Private integration project	Quoted only after scoping	Can data access, integration hours, security work and support be priced sustainably?

4. Unit-economics model

Track actual rather than assumed economics. Useful formulas: Monthly recurring revenue = paying accounts x average monthly price. Gross margin = (revenue - direct delivery cost) / revenue. Contribution per account = price - attributable hosting, API, data and support costs. Payback period = acquisition cost / monthly contribution. A product should not be scaled until real usage, retention and support costs are understood.

5. Roadmap to business evidence

- Phase 1: interview 5-10 potential education/civilian users and document the current workflow before describing a solution.
- Phase 2: deliver one bounded pilot using synthetic or explicitly approved data. Select a measurable baseline metric and test plan.
- Phase 3: compare the same scenario set, record failures and user time, and verify that any claimed benefit reproduces.
- Phase 4: quote pricing to real prospects; track conversion, repeat use, support time and contribution margin.
- Phase 5: only after security and legal review, evaluate hosting, data retention, access control and integration terms.

6. Responsible product boundaries

The current public positioning should be for research/education or an approved generic civilian workflow. Do not claim the prototype is approved by the Ministry of Defence, can be used for operational military decisions, or is validated on military data. A defence procurement business model requires its own authorized customer discovery, compliance pathway and security requirements; none is established here.

References and source notes

[1] Ministry of Civil Aviation - Airports Authority of India
https://www.civilaviation.gov.in/node/4093
Government overview of AAI responsibilities for airports, air navigation services, terminals and cargo facilities.

[2] AAI Civil Aviation Training College - Contact
https://catc.aai.aero/en/contact
Official address and contact information for the Civil Aviation Training College, Prayagraj.

[3] NASA Prognostics Center of Excellence Dataset Repository
https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/
Official NASA repository lists turbofan engine degradation simulation data and the C-MAPSS benchmark. It is not military telemetry.